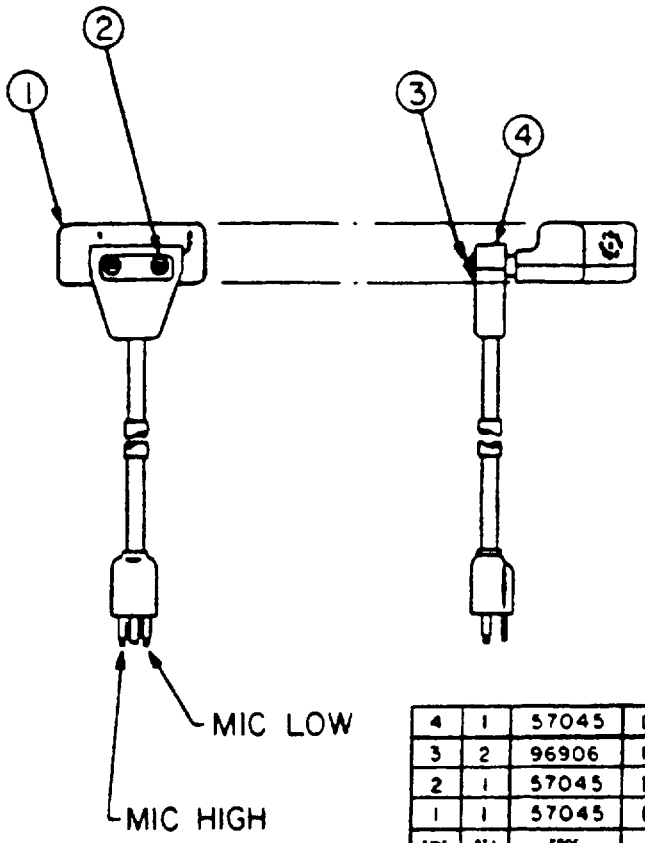
MILITARY SPECIFICATION SHEET

MICROPHONE, LINEAR, M-171/AIC

This specification is approved for use by the Avionics Research and Development Activity of the Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

The complete requirements for acquiring the microphone described herein shall consist of this specification and the latest issue of MIL-M-58109.



4	1	57045	DLD6 001 635	CABLE ASSY
3	2	96906	MS-35233-7	SCREW
2	1	57045	D6 001 640	PLATE
1	1	57045	D6 002 571	MICROPHONE, ELEMENT
FIG ITEM NO	QTY PER NHA	FSCM NO	PART OR IDENTIFYING NO	NOMENCLATURE OR DESCRIPTION

PARTS LIST

FIGURE 1. Microphone assembly.

AMSC N/A

FSC 5965

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REQUIREMENTS:

DRAWINGS: See Figure 1

ARMY

DLD6002570 - Microphone, Mask M-171/AIC Assembly

COMPONENT PARTS: (See Figure 1)

Microphone element: One each, D6002571.
Boom, boom mount and clamp: One each.
Cord assembly = one each DLD6001635.

WEIGHT = The weight of the Microphone Assembly, DLD6002570, shall not exceed 33 grams.

PERFORMANCE:

Microphone amplifier, internal. The internal microphone amplifier shall be capable of accepting impressed power from the intercommunication systems at a maximum supplied power level of 10 V dc at 8 milliamperes (mA). The internal microphone amplifier will reflect sufficient gain to provide a nominal 3.7 millivolts (mV) audio output when connected to a 150 ohm load when subjected to 105 decibels (dB) re: 20 micropascals (uPa), one-fourth inch from the face of the microphone.

Impedance (see 3.7.2.1). The electrical impedance of Microphone M-170/AIC at any frequency over the range of 800 to 3,000 Hz shall be such that the output voltage shall remain 3.7 mV ±3 dB when connected to a 150 ohm ±1 percent load impedance.

Sensitivity at ground level (see 3.7.3.1.1). When measured as specified herein, the sensitivity of the microphone at ground level shall be 3.7 mV ±3 dB. After subjection to any or all environmental tests, sensitivity shall not vary by more than 3 dB from the initial reading.

Noise-immunity characteristic (see 3.7.3.4 and 4.7.4.4). The noise-immunity characteristic of the microphone shall intersect the zero axis at 2,500 Hz minimum, and have a slope of at least 6 dB per octave. See figure 2 as an example of the crossover point ocurring at 2,500 Hz and a falling slope of 6 dB per octave as the frequency decreases.

Frequency response at ground level (see 4.7.4.2.1). The frequency response characteristic of the microphone at ground level, between 400 and 6,000 Hz, shall be within the response envelope shown on figure 3 herein. After subjection to any or all environmental tests, the frequency response shall not vary by more than 3 dB from the initial readings.

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Sensitivity and frequency response characteristic to a close sound source for noise-immunity characteristic measurement. The frequency response characteristics to a close sound source shall be established as previously specified herein. The sensitivity of the microphone to the close sound source specified herein shall be determined by measuring the voltage developed by the microphone across a 150 ohm ±1 percent resistive load at a mean frequency of 1,000 Hz.

Sensitivity and frequency response characteristic to distant sound for noise-immunity characteristic measurements. The frequency response characteristic and sensitivity to a distant sound source shall be determined by measuring the voltage output of the microphone developed across a 150 ohm ±1 percent resistive load when the microphone is placed on axis 1 meter from a sound source in the free field.

Conductor strain relief and twist (see 3.7.4.4). With the pins of one conductor plug rigidly in the microphone and in a vertical position, a 5-pound weight shall be suspended from the pins of the outer connector plug of the cord assembly for a period of at least 30 minutes.

The impedance of the microphone with cord assembly in place shall be measured at 1,000 Hz and a mechanical inspection shall be made before and immediately upon completion of the test. A dielectric withstanding voltage test shall be made on the cord assembly in accordance with the procedure specified in 4.7.5.5 after the impedance measurement and mechanical inspection have been made.

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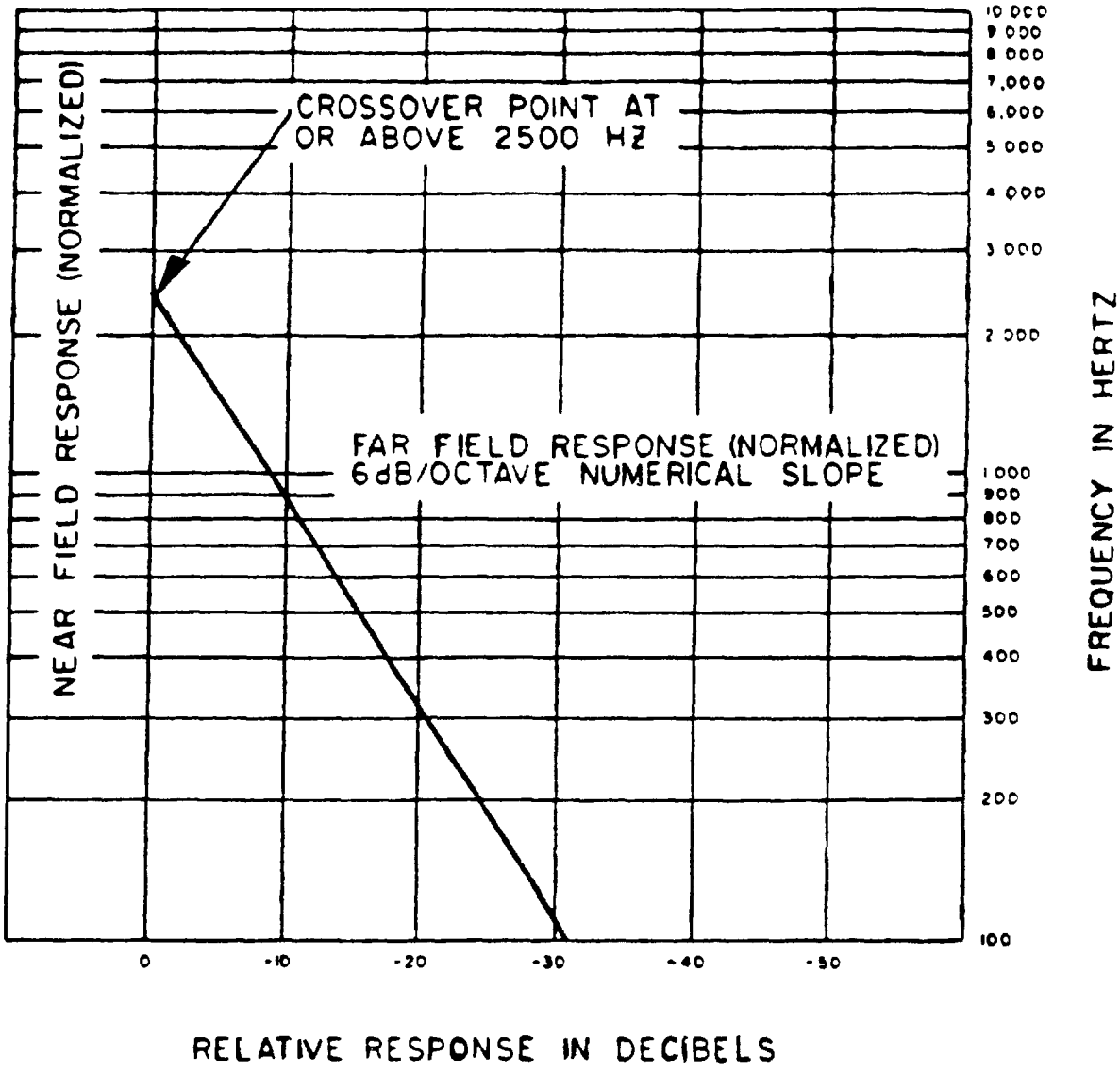
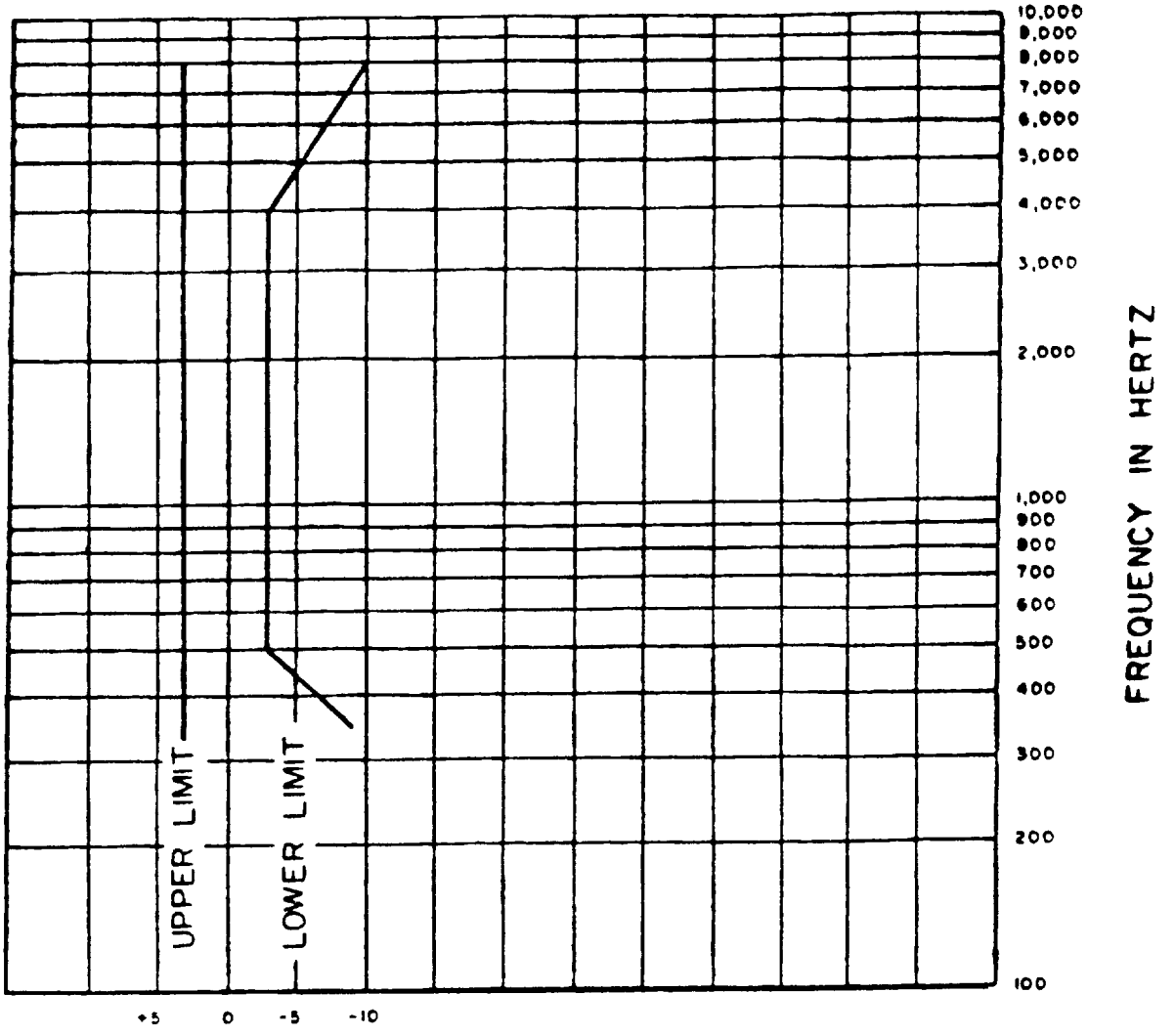


FIGURE 2. Noise-immunity characteristics.

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FREQUENCY IN HERTZ

FIGURE 3. Frequency response - limit curves.

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Tests: The microphone assembly shall be tested in accordance with the tests listed in Table 1, MIL-M-58109(AV).

Impedance (see 3.7.2.1). The microphone shall be placed in front of the sound source as specific in 4.4.1. The microphone shall be supplied with 10 volts dc from a regulated dc power supply (the power supply current shall be maintained at a level not to exceed the maximum value of 8 mA) connected to a load resistance and to a sensitive graphic recorder or sensitive true RMS voltmeter as shown on figure 4. The response of each microphone shall be run with load resistor of 150 ohms, 1 percent at each of the four test frequencies of 800, 1,000, 2,000, and 3,000 Hz.

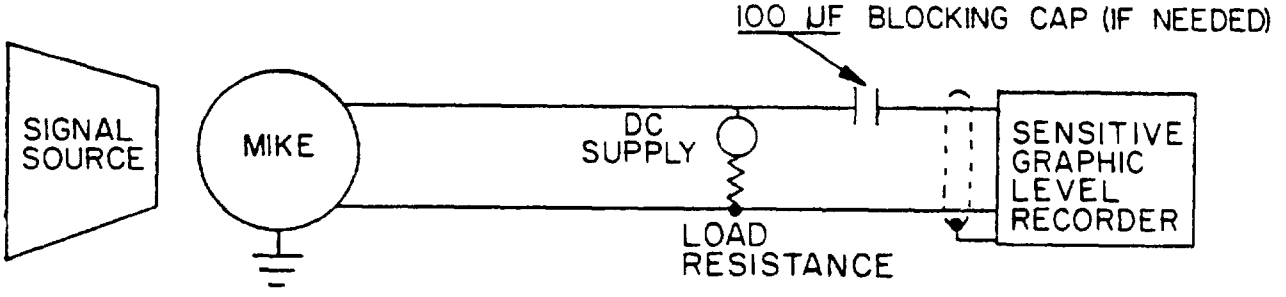


FIGURE 4. Impedance measurement schematic.

Sensitivity to a close sound source. Position the microphone under test with its face one-fourth inch on axis to the driver unit of an artifical voice (B&K type 4219, or equal) with the diaphragm of the microphone parallel to the driver unit. The artifical voice shall have been previously calibrated by means of a condenser microphone (B&K type 4133, or equal, certified traceable to National Bureau of Standards). This calibration will be conducted with an applied constant sound pressure of 3.6 pascals per square centimeter (105 dB sound pressure level, Re: 20 uPa). (A regulated dc power supply shall be used to power the internal microphone amplifier). Determine the microphone output readings using the HP334A Distortion Analyzer, or equal, set for voltage measurement (or a suitable electronic voltmeter, having an input impedance of at least 0.5 megohm connected across a noninductive load resistance of 150 ohms). The sensitivity shall be 3.7 mV ±3 dB at 1,000 Hz.

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Identification of product. The identification marking on the microphone shall include the following information only:

* MICROPHONE, LINEAR M-171/AIC
MANUFACTURER'S NAME OR FSCM
DATE CODE (OPTIONAL)

* Data to be inserted by the contractor per MIL-STD-1285. Marking shall be done in accordance with MIL-STD-1285.

Intended use. Microphone, Linear, M-171/AIC is a noise-cancelling dc powered, high gain microphone designed for use in a pressure type protective mask or where the use of a pressurized helmet is required. The microphone is intended for use with the C-11746/ARC Communication System Control Unit as a part of the airborne communication system which will provide communication of high intelligibility under the extreme noise conditions encountered in military aircraft. The preservation, packing and marking specified herein are intended for direct shipments to the Government. However, this specification may also be used for the preparation of microphones for shipment from the parts contractor to the original equipment manufacturer.

Custodian:
Army - AV

Preparing Activity
Army - AV
(Project 5965-A191

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